

Unit 6 Practice Problems

Right Triangles and Trigonometry Practice

How to use this file: Copy the notes carefully, then cover the worked examples and redo them without looking. For practice files, work first, then check the answer bank. Detailed solutions are included for Unit 3 through Unit 8 practice as requested.

Unit 6 Practice Problems

Directions

Round decimal answers to the nearest tenth unless an exact answer is requested. Work all problems first, then check the answer bank and detailed solutions.

Problems

1. A right triangle has hypotenuse 13 and one leg 5. Find the other leg.
2. A right triangle has legs 9 and 12. Find the hypotenuse.
3. Determine whether 10, 24, 26 form a right triangle.
4. In a $45^\circ - 45^\circ - 90^\circ$ triangle, a leg is 7. Find the hypotenuse.
5. In a $45^\circ - 45^\circ - 90^\circ$ triangle, the hypotenuse is 18. Find each leg exactly.
6. In a $30^\circ - 60^\circ - 90^\circ$ triangle, the short leg is 5. Find the long leg and hypotenuse.
7. In a $30^\circ - 60^\circ - 90^\circ$ triangle, the hypotenuse is 14. Find the short leg and long leg.
8. For an acute angle θ , opposite side 12, hypotenuse 13. Find $\sin \theta$.
9. For an acute angle θ , adjacent side 5, opposite side 12, hypotenuse 13. Find $\cos \theta$ and $\tan \theta$.
10. A right triangle has hypotenuse 20 and an angle 35° . Find the side opposite the angle.
11. A right triangle has angle 40° and opposite side 12. Find the adjacent side.
12. Find an acute angle θ if the opposite side is 7 and the adjacent side is 10.
13. Use the complementary-angle identity to rewrite $\sin 28^\circ$ as a cosine.
14. A 20-ft ladder reaches 18 ft up a wall. Find the angle the ladder makes with the ground.
15. A building is 50 ft tall. From the ground, the angle of elevation to the top is 30° . Find the horizontal distance to the building.
16. Find the area of a triangle with sides 10 and 14 enclosing an angle of 50° .
17. Use Law of Sines: $A = 35^\circ$, $a = 12$, $B = 80^\circ$. Find b .
18. Use Law of Sines: $A = 42^\circ$, $a = 15$, $b = 10$. Find angle B .
19. Use Law of Cosines for sides 7 and 9 with included angle 60° . Find the third side.
20. Use Law of Cosines to find angle C , opposite side 13, when the other sides are 10 and 14.
21. For an SSS non-right triangle, which law should you usually use first?
22. For a triangle where you know ASA or AAS information and need a side, which law should you usually use?
23. Find the diagonal of a rectangle with side lengths 8 and 15.
24. A right triangle has angle 52° and adjacent side 11. Find the hypotenuse.
25. Two perpendicular forces have magnitudes 30 and 40. Find the resultant magnitude.

Answer Bank

1. 12.
2. 15.
3. Yes.
4. $7\sqrt{2}$.
5. $9\sqrt{2}$.
6. Long leg $5\sqrt{3}$, hypotenuse 10.
7. Short leg 7, long leg $7\sqrt{3}$.
8. $\frac{12}{13}$.
9. $\cos \theta = \frac{5}{13}$, $\tan \theta = \frac{12}{5}$.
10. 11.5.
11. 14.3.
12. 35.0° .
13. $\cos 62^\circ$.
14. 64.2° .
15. 86.6 ft.
16. 53.6.
17. 20.6.
18. 26.5° .
19. $\sqrt{67} \approx 8.2$.
20. 63.0° .
21. Law of Cosines.
22. Law of Sines.
23. 17.
24. 17.9.
25. 50.

Detailed Solutions

1. Use $a^2 + b^2 = c^2$. The missing leg is $\sqrt{13^2 - 5^2} = \sqrt{169 - 25} = \sqrt{144} = 12$.
2. The hypotenuse is $\sqrt{9^2 + 12^2} = \sqrt{81 + 144} = \sqrt{225} = 15$.
3. Check $10^2 + 24^2 = 100 + 576 = 676 = 26^2$. Yes, they form a right triangle.
4. In a $45 - 45 - 90$ triangle, hypotenuse $= x\sqrt{2}$. With leg 7, the hypotenuse is $7\sqrt{2}$.
5. If hypotenuse is 18, each leg is $18/\sqrt{2} = 9\sqrt{2}$.
6. In a $30 - 60 - 90$ triangle, if short leg is x , long leg is $x\sqrt{3}$ and hypotenuse is $2x$. With $x = 5$, the answers are $5\sqrt{3}$ and 10.
7. The short leg is half the hypotenuse: $14/2 = 7$. The long leg is $7\sqrt{3}$.
8. Sine is opposite over hypotenuse, so $\sin \theta = \frac{12}{13}$.
9. Cosine is adjacent over hypotenuse: $\frac{5}{13}$. Tangent is opposite over adjacent: $\frac{12}{5}$.
10. Opposite over hypotenuse uses sine: $x = 20 \sin 35^\circ \approx 11.5$.
11. Tangent is opposite over adjacent: $\tan 40^\circ = 12/x$. Thus $x = 12/\tan 40^\circ \approx 14.3$.
12. Use $\tan \theta = 7/10$. Then $\theta = \tan^{-1}(0.7) \approx 35.0^\circ$.
13. Sine of an angle equals cosine of its complement. The complement of 28° is 62° , so $\sin 28^\circ = \cos 62^\circ$.
14. The height is opposite and the ladder is the hypotenuse: $\sin \theta = 18/20$. Therefore $\theta = \sin^{-1}(0.9) \approx 64.2^\circ$.
15. The height is opposite and horizontal distance is adjacent: $\tan 30^\circ = 50/x$. So $x = 50/\tan 30^\circ \approx 86.6$ ft.
16. Use $A = \frac{1}{2}ab \sin C$: $A = \frac{1}{2}(10)(14) \sin 50^\circ = 70 \sin 50^\circ \approx 53.6$.
17. Use $\frac{a}{\sin A} = \frac{b}{\sin B}$: $b = \frac{12 \sin 80^\circ}{\sin 35^\circ} \approx 20.6$.
18. Use $\frac{\sin B}{b} = \frac{\sin A}{a}$. Then $\sin B = \frac{10 \sin 42^\circ}{15}$, so $B \approx 26.5^\circ$.
19. Law of Cosines: $c^2 = 7^2 + 9^2 - 2(7)(9) \cos 60^\circ = 49 + 81 - 63 = 67$. So $c = \sqrt{67} \approx 8.2$.
20. Use $13^2 = 10^2 + 14^2 - 2(10)(14) \cos C$. Then $169 = 296 - 280 \cos C$, so $\cos C = 127/280$. Thus $C \approx 63.0^\circ$.
21. SSS gives three sides and no angles. Law of Cosines is designed to find an angle or side from SSS/SAS information.
22. ASA or AAS gives an angle-side pair or lets you find the third angle, so Law of Sines is usually efficient.
23. The diagonal is the hypotenuse: $\sqrt{8^2 + 15^2} = \sqrt{64 + 225} = 17$.
24. Cosine uses adjacent over hypotenuse: $\cos 52^\circ = 11/x$. Thus $x = 11/\cos 52^\circ \approx 17.9$.
25. Perpendicular forces form a right triangle. Resultant magnitude is $\sqrt{30^2 + 40^2} = 50$.